

Visualizing Light-Induced Microstrain and Phase Transition in Lead-Free Perovskites using Time-Resolved X-Ray Diffraction

Yingqi Wang^{‡§ a}, Cunming Liu^{‡b}, Yang Ren^c, Xiaobing Zuo^b, Sophie E. Canton^d, Kaibo Zheng^e, Kuangda Lu^f, Xujie Lü^a, Wenge Yang^{a*}, Xiaoyi Zhang^{b*}

^aCenter for High Pressure Science & Technology Advanced Research, 1690 Cailun Rd, Pudong, Shanghai, 201203, China

^bX-ray Science Division, Argonne National Laboratory, 9700 S. Cass Avenue, Lemont, Illinois, 60439, United States

^cDepartment of Physics, City University of Hong Kong, 83 Tat Chee Avenue, Kowloon, Hong Kong, China

^dEuropean XFEL, Holzkoppel 4, 22869 Schenefeld, Germany

^eDepartment of Chemical Physics and Nanolund, Lund University, Box 124, 22100 Lund, Sweden

^fBiomedical Engineering Department, Peking University, Beijing, 100871, China

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ABSTRACT: Metal halide perovskites have emerged as promising materials for optoelectronic applications in last decade. A large amount of effort has been paid to investigate the interplay between crystalline lattice and photo-excited charge carriers as it is vital to their optoelectronic performance. Among them, ultrafast laser spectroscopy has been intensively utilized to explore the charge carrier dynamics of perovskites, from which the local structural information can only be extracted indirectly. Here, we have applied time-resolved X-ray diffraction (TR-XRD) technique to interrogate the structural dynamics of a prototypical two-dimensional (2D) lead-free halide perovskite $\text{Cs}_3\text{Bi}_2\text{Br}_9$ nanoparticles across temporal scales from 80 ps to microseconds. We observed a quick recoverable (a few ns) photoinduced microstrains up to 0.15% and a long existing lattice expansion ($\sim$ a few hundred ns) at mild laser fluence. Once the laser flux beyond 1.4 mJ/cm^2 , the microstrain saturates and the crystalline phase partially transfers into a disordered phase. This photoinduced transient structural change can recover within nanosecond time scale. These results indicate that photoexcitation of charge carriers couples with lattice distortion, which fundamentally affects dielectric environment and charge carriers transport.

INTRODUCTION

Metal halide perovskites have recently attracted increasing attention with their superior photovoltaic and optoelectronic properties. Perovskite solar cells have gone through an explosive increase in power conversion efficiency (PCE) from 3.8% to over 25% in the last decade.^{1, 2} They have also been extensively explored for the applications of light-emitting diodes and lasers due to their outstanding emission properties.³⁻⁵ Although significant progress has been made in device efficiency,⁶ the fundamental and comprehensive understanding of their photo-induced structural dynamics, which are directly linked to their performance and stability, are largely underexplored.

The strongly polar and soft fluctuating nature of the metal halide perovskite lattice gives rise to interesting structure-property relationships.⁷⁻¹⁰ Structural rearrangement readily occurs under external stimulation or perturbation, such as photoexcitation, due to lattice softness.¹¹⁻¹⁶ This is in contrast to

classical inorganic semiconductors whose atomic displacement is usually small in similar conditions.¹⁷⁻¹⁸ The dynamic lattice response directly contributes to the dielectric environment experienced by charge carriers in metal halide perovskites, therefore, influences their properties such as exciton binding strength, charge carrier mobility, and recombination rate. Thus, directly unveiling the structural dynamics upon photoexcitation is indispensable for fundamentally understanding the optoelectronic behavior in these perovskite materials.

Time-resolved optical techniques, such as ultrafast Raman, infrared, Terahertz, and visible light spectroscopy, have revealed valuable information on transient phonon and electronic structures of metal halide perovskites upon photoexcitation.^{8, 19-20} Although optical measurements can probe dipole-allowed electronic transitions with high time-resolution, they only indirectly probe the atomic positions, and cannot resolve optical dark states as well as the long-range order of lattice structure. Recently, emerging ultrafast X-ray and

electron scattering techniques have enabled us to directly reveal structural dynamic information of materials upon photoexcitation. Pioneering work applying ultrafast electron or X-ray scattering on lead halide perovskites has revealed light-induced phase transitions²¹ or rotational disordering of the Pb-Br octahedron,²² and a very recent work analyzed the structural distortion with spherical polaronic local strain field.²³ The structural variation in lead-free perovskite is different from lead-containing one. Recent X-ray spectroscopy results suggest the formation of halide dimer (V_k center) under photoexcitation in bismuth halide in bismuth halide perovskite.²⁴ However, the impact of structural distortion on long range order still needs in-depth inspection. Such information is vital for understanding the phonon-carrier interaction and phase stability which directly impacts carrier lifetime and device performance.

Here, we employed ultrafast laserinitiated time-resolved XRD (with time resolution of 79 ps) to directly track the structural evolution of a prototypical 2D lead-free perovskite ($\text{Cs}_3\text{Bi}_2\text{Br}_9$) upon photoexcitation. Light-induced microstrain up to 0.15% emerges with laser power as low as 0.6 mJ/cm^2 , concurrent with a lattice expansion. The microstrain relaxes within ten nanoseconds while the fully recovery of lattice expansion takes hundreds of nanoseconds. When the laser excitation flux exceeded 1.4 mJ/cm^2 , microstrain does not further increase, but a secondary disordered phase appeared. Both the observed microstrain and phase boundary would increase the scattering and recombination probability of the photogenerated carriers, thus impairing the performance. Solving this problem in the design and development of lead-free perovskites could help enhance its performance and stability.

RESULT AND DISCUSSION

TR-XRD patterns under moderate laser excitation power.

$\text{Cs}_3\text{Bi}_2\text{Br}_9$ nanocrystals (NCs) were synthesized by a modified ligand-assisted precipitation method,²⁵ and then dispersed in toluene. Figure S1a shows the transmission electron microscopy (TEM) images of the as-synthesized NCs. The $\text{Cs}_3\text{Bi}_2\text{Br}_9$ nanoparticles are hexagonal plates NCs with average lateral diameter of 80 nm (Figure S1b). The thickness of the nano plates is estimated to be $\sim 20 \text{ nm}$ from vertically

oriented flakes in the TEM images. TR-XRD experiments were performed at the Advanced Photon Source 11-ID-D of the Argonne National Laboratory. A 400 nm, a 110 fs (FWHM) laser “pump” pulse was used to create band-gap excitation of the NCs, followed by 79 ps (FWHM) X-ray “probe” pulses at certain time delays to monitor the structural changes via collecting the X-ray scattering signals (Figure S3). The 400 nm laser is selected, which is just over the exciton peak ($\sim 432 \text{ nm}$, Figure S4), to minimize the undesired thermal and hot carrier effect. The details of the experimental setup can be found in the method section and Supporting Information (SI). Figure 1a shows the static XRD pattern of the $\text{Cs}_3\text{Bi}_2\text{Br}_9$ NC liquid suspension with the scattering contribution from the pure solvent been subtracted. $\text{Cs}_3\text{Bi}_2\text{Br}_9$ (space group $P\bar{3}m1$) is a vacancy-ordered 2D perovskite, formed by corrugated layers of corner-sharing BiBr_6 octahedra (Figure 1b). The position and intensity of $\text{Cs}_3\text{Bi}_2\text{Br}_9$ NCs diffraction peaks match well with those from the calculated XRD pattern of the bulk single crystal analog (ICSD PDF # 70-0493, Figure 1a). This indicates that NCs retain good crystallinity and purity. The strongest diffraction at Q (scattering vector) of $2.22 \text{ \AA}^{-1}$ comes from the (022) crystal plane (Figure 1b).

Figure 1c shows a typical difference spectrum between the XRD data before and 150 ps after laser excitation with a flux of 0.6 mJ/cm^2 . All diffraction peaks show similar behavior upon photoexcitation, with asymmetric intensity reduction at higher Q range and increased intensity in the lower Q range. When zooming into the strongest (022) diffraction peak (Figure 1d), we can clearly observe its shift towards smaller Q and broadening after laser excitation. The data analysis based on PVII peak function (details in SI) reveals a 16% reduction in the peak height, and 14% increase in the peak width.

Previous studies have shown that when semiconducting thin film or metal nanoparticle is excited by photon with energy much larger than its band gap (generating the so-called ‘hot carriers’), the diffraction peak intensity changes due to the laser heating effect can be well described using the Debye-Waller (DW) model.^{22, 26-27} Here, we apply this model to check whether the observed peak broadening and intensity attenuation after photoexcitation are caused by the similar effect:

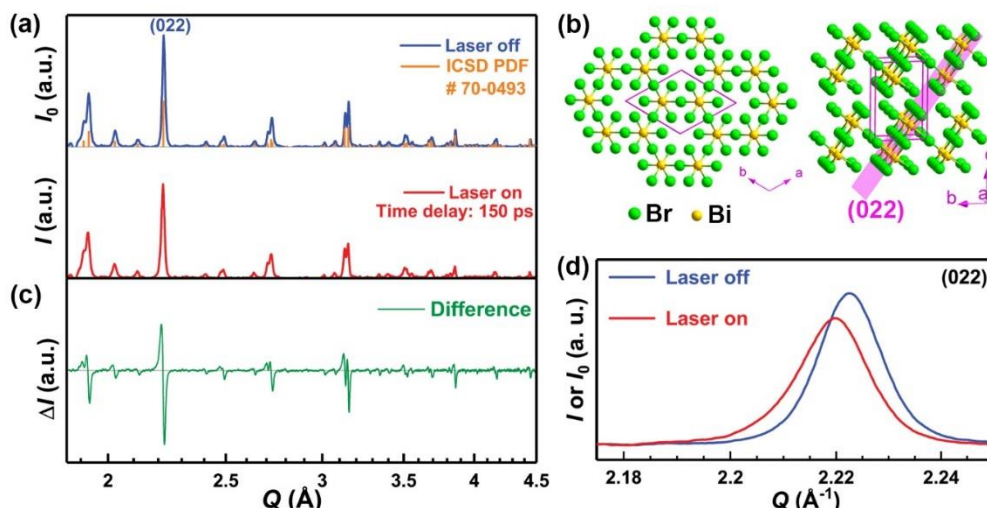


Figure 1. (a) The XRD patterns of the $\text{Cs}_3\text{Bi}_2\text{Br}_9$ NC suspension before (laser off) and 150 ps after (laser on) the photoexcitation, which are compared to the standard pattern of the $\text{Cs}_3\text{Bi}_2\text{Br}_9$ (ICSD PDF# 70-0493). (b) Crystal structure of $\text{Cs}_3\text{Bi}_2\text{Br}_9$ with trigonal $P\bar{3}m1$ space group, viewed from the directions parallel and perpendicular to the $[\text{Bi}_2\text{Br}_9]^{3-}$ layer. (c) Difference of XRD patterns at time delay of 150 ps with pulse energy density of 0.6 mJ/cm^2 ($\Delta I = I(\text{laser on}) - I_0(\text{laser off})$). (d) The comparison of (0 2 2) diffraction peak before and 150 ps after laser photoexcitation.

$$A(Q, t) \propto \exp\left(-\frac{1}{3}Q^2 \langle u^2(t) \rangle\right) \quad (1)$$

where $A(Q, t)$ is the time-dependent integrated peak intensity at the scattering vector Q , $\langle u^2(t) \rangle$ is the mean-square displacement of the photoexcited-atoms averaged spatially over the sample, and $\exp(-Q^2 \langle u^2 \rangle / 3)$ is denoted as DW Factor, or temperature factor. If the thermal effect is dominant, *i.e.* the peak evolution follows the DW model, $-\ln(A/A_0)$ will be proportional to Q^2 (A_0 is the integrated intensity before photo excitation). We fitted the diffraction patterns with a sum of multiple PVII peak functions to obtain peak parameters (Figure 2a). As shown in Figure 2b, the $-\ln(A/A_0)$ is almost constant on Q^2 ranging from 4 to 18 ($\text{\AA}^{-2}$) (light blue line and scatters), which is different to the one predicted by the DW model reflecting a pure thermal expansion trend (dashed pink line). This implies that the thermal effect from laser heating is not the dominate cause of peak intensity change in our studies. Actually, our band-gap photoexcitation should produce mostly cold charge carriers, which could not contribute a large lattice heating effect.^{22, 28}

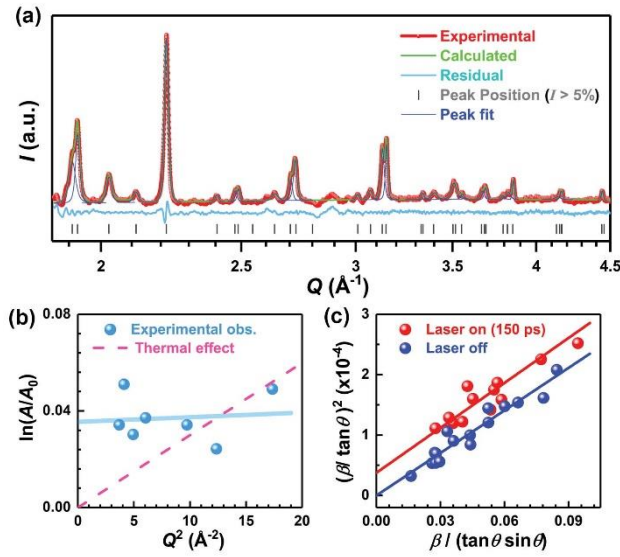


Figure 2. Peak analysis of the $\text{Cs}_3\text{Bi}_2\text{Br}_9$ diffraction pattern 150 ps after laser excitation (400 nm, 0.6 mJ/cm^2). (a) Peak decomposition by profile fitting using the PVII peak function (Red: experimental data, Green: calculated fitting including all fitted peaks; Light gray: calculated Bragg peak position; Blue: fitted peaks; Cyan: fitting residual). (b) Semilog plot of intensity change as a function of the squared scattering factor Q^2 (light blue line: guide for the eye), which is compared to the one deviated from the Debye-Waller model (dashed pink line). (c) Linear fitting of integral-breadth analysis using Halder-Wagner plots before and 150 ps after laser excitation, where the increase of intercept indicates rising microstrain after photoexcitation.

Two other possible explanations of peak broadening are the changes in strain distribution or the incoherent domain size of diffraction.²⁹ These two factors follow different dependences over the diffraction vector, which allows separating the cause of peak broadening over a wide 2θ range. Detailed structural analysis of the observed diffraction data was performed using the Halder-Wagner equation^{29, 30} to further examine the origin of the peak broadening:

$$\left(\frac{\beta}{\tan \theta}\right)^2 = \frac{K\lambda}{D} \cdot \frac{\beta}{\tan \theta \sin \theta} + 16\epsilon^2 \quad (2)$$

Here, the integral breadth β is defined as the peak area divided by peak height, D the coherent crystal domain size, ϵ the microstrain, K a dimensionless shape factor, and λ the X-ray wavelength ($1.0589 \text{\AA}$). Figure 2c shows the Halder-Wagner plot of $\text{Cs}_3\text{Bi}_2\text{Br}_9$ NCs before and 150 ps after laser excitation. The slope and y intercept from the linear fitting correspond to $K\lambda/D$ and $16\epsilon^2$, respectively. Before laser excitation, the strain in the NCs was close to zero. After laser excitation, the fitted line shifts upward while the slope remained unchanged, suggesting an increase in the microstrain without changes in the particle size. The crystal domain size D is calculated to be $\sim 54 \text{ nm}$ from the fitted slope, while the increase in crystal strain ϵ is $\sim 0.15\%$ from the y intercept.

We examined the microscopic scenario to understand the origin of the microstrain increase. Because of the highly deformable and polar nature of the metal halide framework, they are prone to lattice relaxation after electronic excitation.³¹⁻³² This is in contrast to classical crystalline semiconductors in which delocalized electronic states act as Bloch waves, and are comparably insensitive to local lattice dynamics.^{8, 19} Time-resolved optical spectroscopy experiments indicate local geometrical distortion upon photoexcitation.^{8, 19, 33, 34} Ultrafast electronic diffraction and X-ray absorption suggest that the local distortion could be rotational disorder of the halide octahedral, possibly forming small polaron in the configuration halide dimer.^{22, 24} It is concluded that this geometry distortion is driven by the vibrational wave packet in the electronically excited state,^{8, 34} and it completes in 90 fs, which is much faster than the traditional thermal effect.³⁵ In lead halide perovskite, the lattice deformation forms large polarons, which have been proposed to screen the Coulombic potential and protect charge carriers from scattering.^{9, 36} In lead free halide perovskite, there are more reports on small polaron formation from the charge-lattice interaction.^{32, 37} For example, the self-trapped state is believed to be responsible for the broad photoluminescence emission in $\text{Cs}_3\text{M}_2\text{I}_9$ ($M = \text{Bi}, \text{Sb}$).³⁸ Upon photoexcitation, the formation of V_k center (Br_2^- dimer)²⁴ in $\text{Cs}_3\text{Bi}_2\text{Br}_9$ produce an internal electrical field, leading to anomaly in photoconductivity.³⁹ Here, we visualize the increase of microstrain in photoexcited $\text{Cs}_3\text{Bi}_2\text{Br}_9$, which should be a long-range integration of the local distortions. The microstrain could have potential impact on photostability and migration of charge carriers.

TR-XRD kinetics under moderate laser excitation fluence.

Figure 3a shows the difference between the diffraction patterns before and after laser excitation (0.6 mJ/cm^2) at various delays from 150 ps to $4 \mu\text{s}$. Figure 3b is a zoomed-in view of the strongest (022) diffraction peak. Recovery kinetics of the peak height and peak width are similar (Figure 3c top panel), revealing two exponential time components: $\tau_1 = 1.3 \pm 0.2 \text{ ns}$ (72%) and $\tau_2 = 9 \pm 4 \text{ ns}$ (28 %) (See the fitting details in the SI). The kinetic trace of the photo-induced microstrain (Figure 3c, bottom panel) can be fitted well with a single exponential decay of $\tau = 1.5 \pm 0.3 \text{ ns}$, which is consistent with the dominant fast lifetime component (1.3 ns) in peak height and peak width change kinetics. However, the longer one (9 ns) is missing in the microstrain kinetics. This might be because the microstrain extracted based on Eq. (2) has larger uncertainties than that can be resolved from width or height of the strongest peak. The contribution of minor lifetime component is too small to be detected in microstrain kinetics. Importantly, the lifetimes of $\sim 1 \text{ ns}$ and $\sim 9 \text{ ns}$ are comparable to the lifetime in the reported

PL and OTA measurements of $\text{Cs}_3\text{Bi}_2\text{Br}_9$,^{25, 40} suggesting that these structural dynamics component originates from photo-generated excitons.

In Figure 3a and 3b, the negative shift of the diffraction peak position (ΔQ) indicates a lattice expansion taking in place. The kinetics of the Q shift are displayed in Figure 3d. To further understand the photo-induced lattice expansion, we also studied the evolutions of $\Delta a/a$, $\Delta c/c$ and $\Delta V/V$, which are the relative changes of unit cell parameters (Figure 3d). Here, a , c and V were generated from the Le Bail refinement using space group $P-3m$ (164) and PVII peak function. The kinetics of $\Delta a/a$, $\Delta c/c$ and $\Delta V/V$ can be globally fitted using the three-exponential decay function, in which the two short lifetime components ($\tau_1 = 1.3$ ns and $\tau_2 = 9$ ns) are adapted from the kinetics fitting of peak intensity and width changes, plus a long lifetime components 202 ± 32 ns (Figure 3d). The common lifetime components indicate the NC has homogeneous lattice expansion. The photo-generated electrons in the conduction band can populate antibonding states, whereas the holes in the valence band vacate the bonding states, thus, leading to weakening and elongation of the Bi-Br bonds.²⁹ It should be pointed out that the photo-induced unit cell expansion here is distinct from the thermal effect. The NC is surrounded by solvent kept at 18°C (experimental detail in SI). Assuming all the absorbed photon energy converts to heat in the NCs, this would only cause an average 22°C rise in temperature and $\sim 0.22\%$ increase in cell volume at most, much smaller than the 0.43% cell volume expansion we observed (calculation detailed in SI). In addition, the change of diffraction peak does not follow the heat-induced model of Debye-Waller dependence as discussed previously. Therefore, the observed lattice kinetics does not come solely from thermal expansion, but light-induced lattice expansion and microstrain. The small portion of long-lived state ~ 200 ns might originate from the non-radiative recombination of photoinduced charge carriers localized at some trapping states; The lack of this long-lived state in peak width and peak height also indicates that it is not a thermal state.

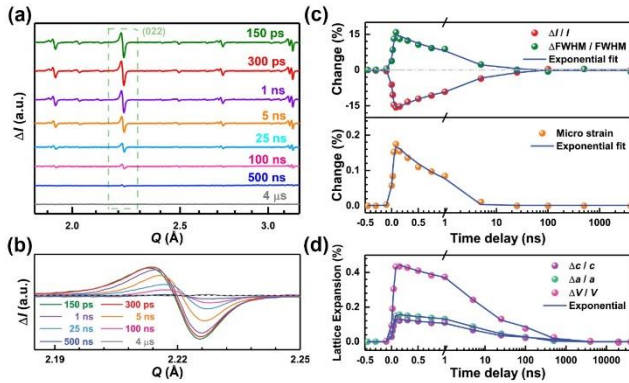


Figure 3. Time resolved analysis of diffraction data under 400 nm excitation at 0.6 mJ/cm². (a) Differential diffraction pattern at time delays from $t = 150$ ps to 4 μ s. (b) Zoom in of the differential data at (0 2 2) plane (Green rectangle in Figure 1a). (c) Time-dependent evolution for the changes in intensity and $FWHM$ of (0 2 2) peak (top panel) as well as microstrain (bottom panel). (d) Time-dependent evolution of lattice expansion examined from the lattice constants a , c and the cell volume V .

TR-XRD under elevated laser excitation fluence.

Figure S7 shows the fluence dependence of the differential TR-XRD signal 300 ps after photoexcitation. By increasing the

laser power, the increasing negative peak on the right side indicates enhanced crystal distortion and loss of peak intensity. Under the low fluence of 0.6 mJ/cm², the sharp positive peak comes from the lattice expansion and shift of diffraction peak towards lower Q . The left side of the positive peak broadened significantly under higher fluence (≥ 1.4 mJ/cm²), suggesting the formation of a disordered phase with larger lattice parameters. The corresponding (022) peak could be fitted with a main peak and a broader side peak at lower Q . It is worth noting that with the increasing of excitation fluence from 1.4 to 5.1 mJ/cm², the position of the light-induced broad side peak remains constant (Figure 4a), suggesting that a metastable state with larger lattice parameters might be produced. Similar analysis was applied to the full range, as shown in Figure 4b and 4c, and a set of side peaks with broader $FWHM$ emerge clearly on the left side of the original crystalline phase. The integrated area change of the main peak and side peak around 2.21 $\text{\AA}^{-1}$ are shown in Figure 4d. Because the decay of the disordered phase correlates with the recovering of the originally phase, the integration area changes of both peaks are fitted with the same double-exponential decay function, revealing two lifetime components 0.25 ± 0.05 ns and 3.2 ± 0.7 ns, respectively. Structural modification on crystallinity state by ultrafast pulse laser has been reported before. A metastable transient phase of 1T-TaS₂ is triggered by infrared laser with optical doping effect.⁴¹ A study on photoexcited VO₂ has attributed the formation of ultrafast disordered phase by transiently modifying the interatomic potential and the atomic positions.⁴² We examined the microstrain condition in the main phase by the Halder-Wagner plot, the same approach used for the analysis of the 0.6 mJ/cm² excitation data (Figure S8). Despite the two-fold increase in the excitation fluence, the photo-induced microstrain in the main phase remained almost unchanged. The microstrain shows saturation at relatively low laser fluence, while the secondary phase keeps growing with increasing fluence. The formation of the secondary disordered phase could be the channel that releases excessive microstrain from the main phase.

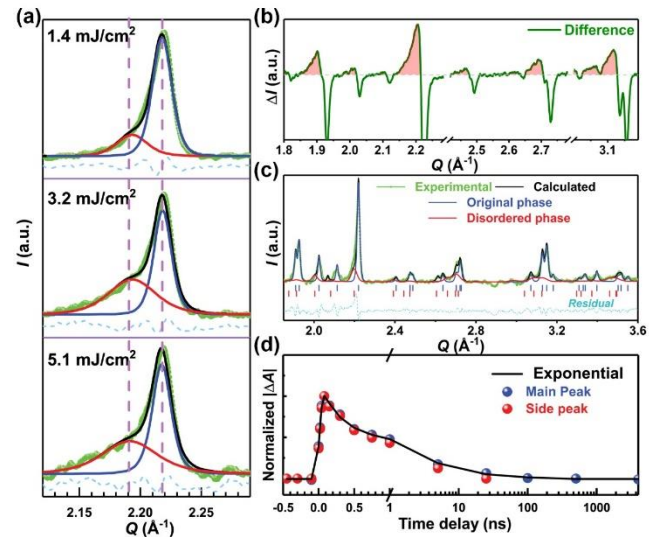


Figure 4. Laser power dependence analysis of diffraction peaks at the 150 ps after photo excitation. (a) fitting of (0 2 2) peak and the emerged side peak representing the 2nd disordered phase under different laser power. With increasing fluence, the position of the side peak remains constant. (b) Red shadow for the 2nd disordered phase in the differential diffraction spectra. (c) A set of side peaks

(red line) with broader *FWHM* emerges on the left side of the original crystalline phase (blue line). (d) Time-dependent integrated area changes for (0 2 2) main peak and its side peak.

It is interesting to compare the recently uncovered structural dynamics of 3D lead halide perovskite CsPbBr_3 with our findings in the 2D lead-free halide perovskite $\text{Cs}_3\text{Bi}_2\text{Br}_9$. CsPbBr_3 is a typical lead halide perovskite where PbBr_6 octahedra connect to each other in a 3D network. Under the photoexcitation, its nanocrystal undergoes an orthorhombic-to-cubic crystalline phase transition which recovers in 510 ps, and no microstrain is observed. Two possible reasons can be ascribed to the differences between CsPbBr_3 and $\text{Cs}_3\text{Bi}_2\text{Br}_9$. First, from the phase diagram and phase transition point of view, CsPbBr_3 has three phases (cubic, tetragonal, and orthorhombic) near room temperature with similar energy, and the phase transitions occur within a small temperature range of 130 °C.⁴³ The photo stimulation could bring in enough energy to drive the phase transitions in this system.⁴⁴ The light-induced local structural distortion or microstrain, if there were any, could be released by crystalline phase transformation. In contrast, the trigonal phase of $\text{Cs}_3\text{Bi}_2\text{Br}_9$ is stable from room temperature to 460 °C (Figure S5). The photo-induced microstrain cannot be released through the phase transition in $\text{Cs}_3\text{Bi}_2\text{Br}_9$ nanocrystals. Second, compared to the CsPbBr_3 with 3D lattice framework, $\text{Cs}_3\text{Bi}_2\text{Br}_9$ in this study has a 2D layer crystalline structure, in which the metal-halide layers are separated by dielectric alkali ions, giving rise to dielectric confinement. The photo-generated charge carriers are more localized and have higher exciton binding energy due to quantum confinement compared to 3D perovskite.^{45, 46} The localized excitons couple more strongly with the lattice in low dimensional perovskites so that self-trap state is widely observed in these materials.³⁸ Thus, the local structural distortion from photoinduced exciton/polaron may lead to an observable increase of microstrain in $\text{Cs}_3\text{Bi}_2\text{Br}_9$.

The structural dynamics of the halide perovskite studied here is intrinsically correlated with the material's properties. The microstrain arises from inhomogeneous variations of crystal lattice that will increase charge carrier scattering, which is directly associated with non-radiative recombination loss.⁴⁷ When the secondary disordered phase emerges under higher excitation fluence, the phase boundary will further increase the carrier scattering and non-radiative loss. Therefore, learning how to reduce and even eliminate the microstrain and secondary phase under light illumination through the design of perovskites is critical for advancing their optoelectronic properties.

CONCLUSION

In summary, the light-induced structural dynamics of a prototypical 2D lead-free perovskite ($\text{Cs}_3\text{Bi}_2\text{Br}_9$) were systematically investigated using time-resolved XRD. Under low excitation fluence of 0.6 mJ/cm², an increase of microstrain was discovered in the NCs due to light-induced local distortion and lattice expansion. As the excitation fluence exceeded 1.4 mJ/cm², a light-induced disordered phase was observed after photoexcitation to release the excessive strain. The light-induced strain and disorder phase in $\text{Cs}_3\text{Bi}_2\text{Br}_9$ recovered in a nanosecond time scale. The photoinduced macrostrain and metastable disordered phases influences the scattering of charge carriers and their dynamics, which may further impact the material properties. In comparison with the recent TRXRD study, where 3D CsPbBr_3 NCs underwent a phase transition upon excitation, the different structural response to release

photo-induced strains is probably attributed to the confined 2D structure of $\text{Cs}_3\text{Bi}_2\text{Br}_9$.

ASSOCIATED CONTENT

Supporting Information: Experimental details, data analysis detail, estimation of heating effect, schematic setup of time-resolved X-ray diffraction, optical absorption spectrum, *in situ* XRD pattern upon heating, lattice expansion under increasing temperature, Differential diffraction intensity 300 ps after excitation, integral-breadth analysis at elevated power density.

AUTHOR INFORMATION

Corresponding Author

* To whom correspondence should be addressed:
xyzhang@anl.gov; yangwg@hpstar.ac.cn.

Present Addresses

§ Department of Physics, University of Washington, Seattle, WA, 98195, United States.

Author Contributions

‡These authors contributed equally.

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